

Kamaljit Aulakh

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PROFESSIONAL SUMMARY

Backend & Platform Engineer with experience building scalable **microservices**, automated **CI/CD** pipelines, distributed backend systems, and developer productivity tooling. Strong foundation in **Java**, **Python**, container orchestration, backend architecture, and test automation at scale. Known for reducing operational bottlenecks, improving system reliability, and delivering high-impact engineering solutions across fintech, enterprise integrations, and platform teams.

EXPERIENCE

Software Engineer II, Vertex Inc - Pennsylvania, USA

June 2025 – Current

- Designed and implemented a fault-tolerant **RESTful API** for an enterprise tax-content management system using **Python (FastAPI)**, **PostgreSQL**, **Redis**, and **Pydantic**, optimizing tax-ID record retrieval.
- Accelerated CI pipelines by 50% by migrating legacy Jenkins groovy based workflows to **GitHub Actions**, **Docker**, and **Azure Container Registry**, leveraging parallelized PyTest execution and matrix workflows with **Datadog** dashboards plotting custom metrics (duration, failure rates, flaky tests).
- Built an AI-augmented failure-analysis workflow using **Azure OpenAI** with a **lightweight RAG pipeline**; implemented **NumPy** based semantic search for structured CSV data and a vector-indexed retrieval system for PDFs (**embeddings + FAISS/Azure AI Search**) to extract failure context and root causes, reducing triage time from **1 full day to <2 hours**.

Web and Data Applications Developer, Open Source Program Office - New York, USA

October 2024 – May 2025

- Developed an open-source research workflow **plugin (PHP/WordPress)** used by 200+ researchers, replacing brittle legacy processes and enforcing **accessibility (WAVE)**, significantly improving data-exploration efficiency.
- Built a **JavaScript-based** inter-repository data transfer system bridging **HEAL and QDR** (Qualitative Data Repository), facilitating secure research data movement and validation; solution adopted and adapted by **Harvard**, demonstrating real-world impact.

Global Integrations Team Summer Intern, Capri Holdings (Michael Kors) - New Jersey, USA

June 2024 – August 2024

- Developed a secure internal audit-monitoring platform using **Node.js (Express)** and **Azure Tables** with **RBAC**, **JWTs**, and refresh-token rotation.
- Streamlined a CDC pipeline by integrating **Debezium SQL connector** into **Spring Boot** with custom **JOLT JSON** validation and transformation, enabling clean migration from legacy tooling to an event-driven architecture.

Software Engineer, IDFC FIRST Bank Ltd, Bangalore, India

June 2022 – August 2023

- Developed high-performance **RESTful** services using **Java Micronaut** and **Python Flask** for mobile and web banking, contributing to a 10% increase in online zero-balance account openings.
- Led a team of 4 engineers to build a **React.js admin panel** with responsive, reusable components, visualizing **Aerospike** data and generating analytical reports via **React Google Charts**.
- Implemented TDD (**JUnit**) achieving 90% coverage, integrated **SonarQube** for code quality, and optimized system performance with **JMeter**, monitored via **Prometheus + Grafana dashboards**.
- Containerized microservices with **Docker + Helm** and deployed on **Kubernetes** through **GoCD** CI/CD pipelines, achieving 98% uptime in production.

Systems Engineer, Tata Consultancy Services, Thane, India

Oct 2021 – June 2022

- Worked in an agile team writing backend logic across 10+ **microservices** using J2EE, **Spring MVC**, and **Spring Boot**, enhancing scalability and performance.
- Implemented front end features with **ReactJS**, **Redux** to integrate security enhancements (behavioral biometrics solution - BioCatch), leading to a 37% improvement in bot detection accuracy and reducing false positives by 20%.

EDUCATION

Syracuse University, College of Engineering and Computer Science, Syracuse, NY

August 2023 – May 2025

Master of Science in Computer Science

GPA: 3.90

Data Structures and Algorithms, Computer Security, Object Oriented Design

Mumbai University, Ramrao Adik Institute of Technology, Mumbai, India

August 2017 – June 2021

Bachelor of Engineering in Information Technology

GPA: 3.74

Web application development, Android Programming, Database Management Systems

SKILLS

Programming & Frameworks: Java, Python, JavaScript, Node.js, FastAPI, Flask, Micronaut, Spring Boot, React.js

Databases & Caching: PostgreSQL, Aerospike, Redis, MongoDB, SQL

Cloud & DevOps: Docker, Kubernetes, Helm Charts, GoCD, GitHub Actions, Azure, CI/CD pipelines

Monitoring & Testing: Prometheus, Grafana, Datadog, JMeter, PyTest, JUnit, SonarQube

Tools: Unix/Linux, Git, Postman, Jira, Confluence, Elasticsearch